Survival-supervised loadings under $\eta = (YF)\tilde{\beta}$: projection scale, notation and the exact F update

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This note settles the definition of the program scores Z_F and their subscripts, and corrects the survival term of the $q(F)$ update in `qF_supervised/qF_supervised_derivation.tex` (equation `eq:BFsurv`). It describes `fit_cox_on_yf(canonicalize = TRUE)`. The earlier configuration (`canonicalize = FALSE`, α_F) is unchanged and is described by the earlier derivations.

1 Notation

- $Y \in \mathbb{R}^{n\times p}$: expression, rows i = patients, columns j = genes; z-standardized per platform
- $L \in \mathbb{R}^{n\times K}$, $F \in \mathbb{R}^{p\times K}$: scores and loadings, $Y \approx LF^\top$; columns indexed by program k
- $Z_F = YF \in \mathbb{R}^{n\times K}$: program scores (projections); entry $z_{ik} = \sum_j y_{ij} f_{jk}$, column $z_{\cdot k} = Y f_{\cdot k}$
- $\tilde{\beta} \in \mathbb{R}^K$: survival coefficients; $\eta_i = \sum_k z_{ik}\tilde{\beta}_k$
- u_i, w_i: score and diagonal curvature of the Cox partial log-likelihood at the current η
- $\zeta_i = \eta_i + u_i/w_i$: working response
- $\zeta_{-k,i} = \zeta_i - \sum_{k'\neq k} \bar{z}_{ik'}\bar{\beta}_{k'}$: working response with every program except k removed

Bars denote posterior means under q: $\bar{f}_{jk} = \mathbb{E}_q f_{jk}$, $\bar{z}_{ik} = \sum_j y_{ij}\bar{f}_{jk}$. Subscripts are always (patient i, gene j, program k). Earlier documents wrote $\mathbf{Z}_F$, "ZF" and $(z_F)_{ik}$ for the same matrix.

2 Scale of the projection

$Y \approx LF^\top$ and $\eta = YF\tilde{\beta}$ are unchanged by $f_{\cdot k} \to f_{\cdot k}/c_k$, $l_{\cdot k} \to c_k l_{\cdot k}$, $\tilde{\beta}_k \to c_k\tilde{\beta}_k$ for any $c_k > 0$. The reconstruction term does not fix c_k, but the survival term depends on it, which is the feedback loop behind the April instability. The fitter fixes the scale by

$$\mathrm{sd}_i(\bar{z}_{ik}) = 1 \quad \text{on the training data,}$$

applied after initialization and at the end of every iteration, with second moments scaled by $c_k^{\mp 2}$ and any loading prior held across sweeps transformed with f (Laplace or exponential rate $\times c_k$, normal variance $/c_k^2$). The ebnm priors on L and $\tilde{\beta}$ are refit at every update and are scale families, so they need no transformation. Consequences:

- $\tilde{\beta}_k$ is a log hazard ratio per training standard deviation of program k's score.
- Prediction for a new patient is $\hat{\eta} = y^\top \bar{F} \bar{\tilde{\beta}}$ with the training $\bar{F}$; no further normalization. (The earlier configuration used $Y\bar{F}_{\cdot k}/\|\bar{F}_{\cdot k}\|$ inside the Cox term while its F update was derived for $Y\bar{F}_{\cdot k}$, so the survival pseudo-observation was on a different scale from the reconstruction one.)
- A program whose score has collapsed (sd $< 10^{-6}$) is left unscaled.

3 The $q(f_{jk})$ update

The Taylor-approximated Cox term is $-\frac{1}{2}\sum_i w_i(\zeta_i - \eta_i)^2$. Write $\eta_i = \eta_{-k,i} + \tilde{\beta}_k(y_{ij}f_{jk} + \sum_{j' \neq j} y_{ij'} f_{j'k})$. Under the mean-field q, the terms in f_{jk} give, with survival weight w_S ($w_S = 1$ is the plain posterior),

$$A_{jk} = \tau_j \sum_i \mathbb{E} l_{ik}^2 + w_S\, \mathbb{E}\tilde{\beta}_k^2 \sum_i w_i y_{ij}^2,$$

$$B_{jk} = \tau_j \sum_i r_{-k,ij}\bar{l}_{ik} + w_S\Big[\bar{\beta}_k \sum_i w_i y_{ij}\zeta_{-k,i} - \mathbb{E}\tilde{\beta}_k^2 \sum_i w_i y_{ij}(\bar{z}_{ik} - y_{ij}\bar{f}_{jk})\Big],$$

and $q(f_{jk})$ is the posterior under the loading prior g_k for one observation $x = B/A$ with variance $1/A$. Here r_{-k} is the reconstruction residual without program k.

Correction. `eq:BFsurv` of the earlier derivation omits the last term, $-\mathbb{E}\tilde{\beta}_k^2 \sum_i w_i y_{ij}(\bar{z}_{ik} - y_{ij}\bar{f}_{jk})$: it removes the other programs from η but not the other genes of program k. Each gene then regresses the whole program-k residual on itself, and with correlated genes the update overshoots by roughly the number of correlated genes. The derivation's own definition of $\eta_{\text{no-}j}$ includes those genes; the error is in the step to $z_{\text{no-}k}$. The configurations behind the reported single-modality results use $\alpha_F = 0$, where the survival term is absent, so they are unaffected; the 2026-08-20 α_F grid used the incomplete term.

Because B_{jk} depends on $\bar{f}_{j'k}$ for $j' \neq j$, the update is a sequential sweep over genes, refreshing $\bar{z}_{\cdot k}$ and $\mathrm{Var}(z_{\cdot k})$ after each gene, with g_k held fixed during the sweep and refit by marginal likelihood from the sweep's $(x, 1/A)$ afterwards. This is the update the multimodal model uses (`multimodal_yfb_update_F_mk()`); the single-modality fitter calls it with working weights $w_S w_i$ and $w_S w_i \zeta_{-k,i}$. At convergence of the sweep with fixed $\tilde{\beta}$ and negligible τ it solves the weighted least squares problem $\tilde{\beta}_k Y f_{\cdot k} \approx \zeta_{-k}$ (test Supervision-T8).

A program with $\max_j A_{jk} < 10^{-8}$ carries no information from either likelihood. With g_k held fixed its posterior would be the prior (zero mean, positive second moment), which revives $l_{\cdot k}$ and $\tilde{\beta}_k$ from nothing; the fitter sets $f_{\cdot k} = 0$ with a point-mass prior instead, which is equivalent to pruning it while keeping K columns.

4 $q(\tilde{\beta}_k)$ and the monitored objective

The coefficient update uses $\mathbb{E}z_{ik}^2 = \bar{z}_{ik}^2 + \sum_j y_{ij}^2 \mathrm{Var}(f_{jk})$: $A_{\beta,k} = w_S \sum_i w_i\, \mathbb{E}z_{ik}^2$, $B_{\beta,k} = w_S \sum_i w_i \bar{z}_{ik}\zeta_{-k,i}$. The monitored objective is

$$\mathbb{E}_q \log p(Y \mid L, F, \tau) + w_S\, \hat{\ell}_{\text{Cox}} - \text{KL terms},$$

with the Cox term re-expanded at the end of each iteration at the current $\bar{\eta}$ and including $\mathrm{Var}_q(\eta_i)$. It carries the same weight w_S as every update, which removes the mismatch between the monitored quantity and the updates in the earlier configuration (survival weighted $\alpha p/(1-\alpha)$ in the monitor against $\alpha_F/(1-\alpha_F)$ in the F update). The Cox Taylor approximation and the per-update prior refits mean the objective is not guaranteed to increase at every iteration.